Initial system

$$\begin{cases} \dot{x}_1 = u \\ \dot{x}_2 = x_1^3 + x_1^2 \sin(t) \end{cases}$$

Series

$$x^0 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \xi_0 + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \xi_{02} + \begin{bmatrix} 0 \\ 6 \end{bmatrix} \xi_{001}$$

Basis Lie elements and their coefficients

$$\begin{aligned}
&\mathcal{A}_1 : g_1 = \xi_0 && v(g_1) = \begin{bmatrix} -1 & 0 \end{bmatrix} \\
&\mathcal{A}_2 : g_2 = \xi_1 && v(g_2) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\mathcal{A}_3 : g_3 = \xi_2 && v(g_3) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\quad g_4 = [\xi_0, \xi_1] && v(g_4) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\mathcal{A}_4 : g_5 = \xi_3 && v(g_5) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\quad g_6 = [\xi_0, \xi_2] && v(g_6) = \begin{bmatrix} 0 & -1 \end{bmatrix} \\
&\quad g_7 = [[\xi_0, \xi_1], \xi_0] && v(g_7) = \begin{bmatrix} 0 & -6 \end{bmatrix}
\end{aligned}$$

Lie elements for principal part

$$\begin{aligned}
&\mathcal{A}_1 : \ell_1 = \xi_0 \\
&\mathcal{A}_4 : \ell_2 = [\xi_0, \xi_2]
\end{aligned}$$

The generating set of the Right ideal

$$\begin{aligned}
&\mathcal{A}_2 : d_1 = \xi_1 && v(d_1) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\mathcal{A}_3 : d_2 = \xi_2 && v(d_2) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\quad d_3 = [\xi_0, \xi_1] && v(d_3) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\mathcal{A}_4 : d_4 = \xi_3 && v(d_4) = \begin{bmatrix} 0 & 0 \end{bmatrix} \\
&\quad d_5 = -6[\xi_0, \xi_2] + [[\xi_0, \xi_1], \xi_0] && v(d_5) = \begin{bmatrix} 0 & 0 \end{bmatrix}
\end{aligned}$$

Principal part

$$\begin{aligned}
&\mathcal{A}_1 : \tilde{\ell}_1 = \xi_0 \\
&\mathcal{A}_4 : \tilde{\ell}_2 = \frac{\xi_{02}}{37} - \frac{6\xi_{001}}{37}
\end{aligned}$$

Approximating system

$$\begin{cases} \dot{x}_1 = -u \\ \dot{x}_2 = \left(-\frac{t^2 x_1}{37} + \frac{3t x_1^2}{37} \right) u \end{cases}$$

Stationary approximating system

Stationary approximation does not exist